HR Employee Attrition Analysis

Workforce Risk & Retention Report

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Dataset: 1,470 employees | Tools: Excel (pivot tables, COUNTIFS, dashboard)
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1. Problem Statement & Approach

Employee attrition is one of the most costly and disruptive challenges any organisation faces. Replacing a single employee can cost 50–200% of their annual salary when recruitment, onboarding, and lost productivity are accounted for. This analysis examines a 1,470-employee dataset to answer a central question: who is leaving, why, and what can the organisation do about it?

The approach followed the CRISP-DM framework — defining the business question, exploring the dataset, cleaning inconsistencies (standardising department labels, encoding attrition as binary), and building pivot tables and calculated fields for attrition rate by segment. Findings are organised around three pillars:

- Departmental and role-level attrition trends
- Demographic risk factors (age, income, marital status, education)
- Satisfaction and performance linkage to attrition

2. Overall Attrition Position

Total Employees	Employees Who Left	Overall Attrition Rate	Avg Monthly Income
1,470	237	16.1%	$6,503

> *At 16.1%, the overall attrition rate exceeds the 10–13% industry benchmark for professional services. The organisation is losing roughly 1 in 6 employees — a signal that systemic retention issues exist across multiple dimensions.*

3. Attrition Trends

3.1 By Department

Human Resources and Sales are the two highest-risk departments, both exceeding 20% attrition — significantly above the company average. Research & Development, despite being the largest department, holds a more moderate rate.

Department	Employees	Attrition Count	Attrition Rate
Human Resources	64	13	20.3%
Sales	446	92	20.6%
Research & Development	960	132	13.8%
Grand Total	1,470	237	16.1%

> *R&D's lower rate (13.8%) is positive given it accounts for 65% of the workforce. However, Sales Representative has a 39.8% role-level attrition rate within Sales — the highest of any specific role in the dataset.*

3.2 By Job Role

Laboratory Technicians and Sales Representatives have the highest absolute attrition counts. When measured by rate, Sales Representatives and Human Resources roles are the most at-risk positions.

Job Role	Attrition Count	Attrition Rate
Laboratory Technician	62	23.9%
Sales Executive	57	17.5%
Research Scientist	47	16.1%
Sales Representative	33	39.8%
Human Resources Specialist	12	23.1%
Manufacturing Director	10	6.9%
Healthcare Representative	9	6.9%
Manager	5	4.9%
Research Director	2	2.5%

3.3 By Tenure

Attrition is sharpest in the first year of employment, with a 34.9% rate among employees with 0–1 years of tenure. It declines steadily thereafter, suggesting that onboarding quality and early engagement are critical retention windows.

Years at Company	Attrition Rate
0–1 years	34.9%
2–3 years	18.4%
4–5 years	13.1%
6–10 years	12.3%

Years at Company	Attrition Rate
11–15 years	6.5%
16+ years	9.4%

4. Demographic Analysis

Attrition does not affect all employees equally. Certain demographic groups show markedly higher rates, pointing to structural or environmental factors beyond individual choice.

4.1 Age Group

Age Group	Employees	Attrition Count	Attrition Rate
18–22	57	27	47.4%
23–27	153	32	20.9%
38–42	881	134	15.2%
43–47	163	17	10.4%
48–52	110	14	12.7%
53+	106	13	12.3%

> *Employees aged 18–22 have a 47.4% attrition rate — nearly 1 in 2 leave. Combined with the 0–1 year tenure finding, this points to an acute early-career retention problem.*

4.2 Marital Status

Marital Status	Employees	Attrition Count	Attrition Rate
Single	470	120	25.5%
Married	674	84	12.5%
Divorced	326	33	10.1%

> *Single employees leave at more than twice the rate of divorced employees (25.5% vs 10.1%), likely because fewer personal ties reduce the cost of switching employers.*

4.3 Income Band

Income is the strongest individual predictor of attrition in this dataset. Employees earning below $2,500/month leave at nearly three times the company average.

Monthly Income Band	Employees	Attrition Count	Attrition Rate
$0 – $2,500	226	77	34.1%
$2,501 – $5,000	523	86	16.4%
$5,001 – $7,500	310	30	9.7%
$7,501 – $10,000	130	19	14.6%
$10,001 – $15,000	148	20	13.5%
$15,000+	133	5	3.8%

> *The attrition rate falls dramatically and consistently as income rises — from 34.1% at the lowest band to just 3.8% at $15,000+. Compensation is the most statistically significant attrition lever in this dataset.*

4.4 Education Field

Education Field	Employees	Attrition Count	Attrition Rate
Human Resources	27	7	25.9%
Marketing	160	36	22.5%
Technical Degree	131	31	23.7%
Life Sciences	606	89	14.7%
Medical	464	63	13.6%
Other	82	11	13.4%

5. Satisfaction Analysis

Employees were rated on Job Satisfaction across four bands. The relationship between satisfaction and attrition is clear and directionally consistent — employees who report low satisfaction leave at double the rate of highly satisfied peers.

5.1 Job Satisfaction

Satisfaction Band	Employees	Attrition Count	Attrition Rate
Low	289	66	22.8%
Medium-Low	280	46	16.4%
Medium-High	442	73	16.5%
High	459	52	11.3%

> *Employees with low job satisfaction leave at 22.8% — double the 11.3% rate among highly satisfied employees. The average satisfaction score across the dataset is 2.73 out of 4, suggesting middling satisfaction company-wide — there is room to improve the median, not just the tail.*

5.2 Business Travel

Travel Frequency	Employees	Attrition Count	Attrition Rate
Travel Frequently	278	69	24.8%
Travel Rarely	1,042	156	15.0%
Non-Travel	150	12	8.0%

> *Frequent business travellers have a 24.8% attrition rate — three times the rate of non-travellers (8.0%). Travel burden is a significant and often underestimated satisfaction driver, particularly among younger employees.*

6. Performance & Attrition

Performance ratings in this dataset are narrowly distributed: nearly all employees are rated at level 3 (Excellent) or 4 (Outstanding), with no employees at levels 1 or 2. This ceiling effect limits the diagnostic power of performance data as a standalone attrition predictor.

Performance Rating	Employees	Attrition Count	Attrition Rate
3 – Excellent	1,244	200	16.1%
4 – Outstanding	226	37	16.4%

> *Performance rating alone does not predict attrition — the rates are nearly identical across both bands (16.1% vs 16.4%). The organisation is losing both average and high performers at similar rates. This is a structural issue affecting all performance levels, not a talent quality problem.*

> *The more significant performance-related risk is concentration of attrition in overtime workers. Employees required to work overtime have substantially higher attrition, affecting both high and average performers. Overwork appears to be a stronger attrition driver than the formal performance rating.*

7. Recommendations

The findings point to four structural levers the organisation can pull. Actions are sequenced by estimated impact and organised by responsible function.

1. **Address compensation at the bottom of the income distribution.** Employees earning below $2,500/month leave at 34.1% — more than double the average. A targeted review of entry-level and junior role salaries, particularly in Sales Representative and Laboratory Technician roles, would directly address the most statistically significant attrition driver. Owner: HR and Finance. Target: close the gap within one annual compensation review cycle.

2. **Build a structured 90-day onboarding and check-in programme.** With 34.9% attrition in year one and 47.4% among employees aged 18–22, early tenure is the highest-risk window. Structured mentorship, role clarity sessions, and 30/60/90-day manager check-ins would significantly reduce early exits. Owner: HR Operations. Target: pilot with new Sales and HR hires within one quarter.

3. **Conduct a workload and overtime audit in high-attrition departments.** The travel and overtime signals suggest that workload and flexibility are driving dissatisfaction more than formal scores alone capture. An anonymous workload survey — specifically in HR and Sales, where attrition exceeds 20% — would surface operational pressure points. Owner: Department heads and HR Business Partners. Target: survey fielded within 6 weeks.

4. **Introduce career pathing for single employees and early-career staff.** Single employees leave at 2.5x the rate of divorced employees. Structured career development conversations, internal mobility programmes, and visible growth trajectories would give this cohort a stronger reason to stay. Owner: L&D and line managers. Target: integrate into the annual performance cycle.

5. **Deploy a targeted retention strategy for Technical Degree and Marketing graduates.** Attrition rates of 23.7% (Technical Degree) and 22.5% (Marketing) suggest these hires may not be finding role-fit or growth opportunity. Exit interview analysis for these cohorts would sharpen the diagnosis. Owner: Talent Acquisition and HR Analytics. Target: pull and analyse existing exit data within one month.

8. Conclusion

The headline attrition rate of 16.1% masks a more concentrated problem: the organisation is disproportionately losing young, early-tenure, lower-income, single employees in HR and Sales — and losing them fast, often within their first year. Meanwhile, performance ratings do little to differentiate who stays and who goes, confirming that this is a structural and environmental issue, not a talent quality issue.

The most actionable levers are compensation at the entry level, onboarding quality in the first 90 days, and workload management in the two highest-attrition departments. Addressing these three areas first — before broader culture or engagement initiatives — will have the most measurable and rapid impact on retention rates.

> *High revenue does not always mean high profit — and high headcount does not mean high retention. The data shows that small, targeted interventions in compensation and onboarding carry more weight than company-wide engagement surveys alone.*